

Assignment Scope: **lectures 4 - 8**

Maximum Grade: 5

Student Name:

Student ID:

Assignment Rules & Regulations:

- Plagiarism, cheating, or any form of academic dishonesty is **strictly prohibited**. All work submitted must be the student's own, and all external sources, such as code snippets, libraries, or academic papers, must be properly cited.
- Requests for extensions must be requested **in advance of the deadline** and should be supported by valid and documented reasons, such as medical or family emergencies. Extensions will be granted at the instructor's discretion.
- If you encounter any issues or require clarifications related to the assignment, promptly communicate with the instructor or teaching assistant. **Do not share your assignment solutions with classmates before the submission deadline.**

Assignment Objectives:

- Topic#1: Fitting models to data.
- Topic#2: Counting basics.
- Topic#3: Discrete probability distributions.
- Topic#4: The normal distribution and central limit theorem.
- Topic#5: Confidence intervals and hypothesis testing.

Assignment Exercises:

Exercise 1:

The number of transistors on Intel central processing units since the early 1970s is given in the table that follows. Fit the model $y = c_1 e^{c_2 t}$ to the data.

CPU	year	transistors
4004	1971	2,250
8008	1972	2,500
8080	1974	5,000
8086	1978	29,000
286	1982	120,000
386	1985	275,000
486	1989	1,180,000
Pentium	1993	3,100,000
Pentium II	1997	7,500,000
Pentium III	1999	24,000,000
Pentium 4	2000	42,000,000
Itanium	2002	220,000,000
Itanium 2	2003	410,000,000

Exercise 2: There are 18 mathematics majors and 325 computer science majors at a college.

- In how many ways can two representatives be picked so that one is a mathematics major and the other is a computer science major?
- In how many ways can one representative be picked who is either a mathematics major or a computer science major?



Exercise 3: Suppose that a password for a computer system must have at least 8, but no more than 12, characters, where each character in the password is a lowercase English letter, an uppercase English letter, a digit, or one of the six special characters *, >, <, !, +, and =.

- How many different passwords are available for this computer system?
- How many of these passwords contain at least one occurrence of at least one of the six special characters?

Exercise 4: How many subsets with more than two elements does a set with 100 elements have?

Exercise 5: Find the probability that if 5 different-sized washers are arranged in a row, they will be arranged in order of size.

Exercise 6: A drawer contains 11 identical red socks and 8 identical black socks. Suppose that you choose 2 socks at random in the dark;

- What is the probability that you get a pair of red socks?
- What is the probability that you get a pair of black socks?
- What is the probability that you get 2 unmatched socks?
- Where did the other red sock go?

Exercise 7: From past experience, a company found that in cartons of DVDs, 90% contain no defective DVDs, 5% contain one defective DVD, 3% contain two defective DVDs, and 2% contain three defective DVDs. Find the mean, variance, and standard deviation for the number of defective DVDs.

Exercise 8:

The probability that a cellular phone company kiosk sells X number of new phone contracts per day is shown below. Find the mean, variance, and standard deviation for this probability distribution.

X	4	5	6	8	10
$P(X)$	0.4	0.3	0.1	0.15	0.05

What is the probability that they will sell 6 or more contracts three days in a row?

Exercise 9: Complete the following probability distribution if $P(6)$ equals two-thirds of $P(4)$. Then find the mean and the standard deviation for the distribution.

X	1	2	4	6	9
$P(X)$	0.23	0.18	?	?	0.015

Exercise 10: Which of the following are binomial experiments or can be reduced to binomial experiments?

- Surveying 100 people to determine if they like Sudsy Soap.
- Tossing a coin 100 times to see how many heads occur.
- Drawing a card with replacement from a deck and getting a heart.
- Testing four different brands of aspirin to see which brands are effective.



Exercise 11: It is reported that 77% of workers aged 16 and over drive to work alone. Choose 8 workers at random. Find the probability that

- a. All drive to work alone. b. More than one-half drive to work alone. c. Exactly 3 drive to work alone.

Exercise 12: R. H. Bruskin Associates Market Research found that 40% of Americans do not think that having a college education is important to succeed in the business world. If a random sample of 5 Americans is selected, find these probabilities.

- a. Exactly 2 people will agree with that statement. b. At most 3 people will agree with that statement.
c. At least 2 people will agree with that statement. d. Fewer than 3 people will agree with that statement.

Exercise 13: The average number of phone inquiries per day at the poison control center is 4. Given that those inquiries occur randomly at the center; find the probability it will receive 5 calls on a given day.

Exercise 14: A person works in a large home improvement store and approaches customers to tell them

about the store's carpet sale. He then asks them if they would like to talk to a sales representative. From past experience, the person has found that the probability of getting a "yes" is about 0.32. Find the probability that the person's first "yes" will occur with the fifth customer.

Exercise 15: Find the z value to the left of the mean so that

- a. 98.87% of the area under the distribution curve lies to the right of it.
b. 82.12% of the area under the distribution curve lies to the right of it.
c. 60.64% of the area under the distribution curve lies to the right of it.

Exercise 16: Find two z values, one positive and one negative, that are equidistant from the mean so that the areas in the two tails add to the following values.

- a. 5% b. 10% c. 1%

Exercise 17: The average number of calories in a 1.5-ounce chocolate bar is 225. Suppose that the distribution of calories is approximately normal with $\sigma = 10$ calories. Find the probability that a randomly selected chocolate bar has:

- a. between 200 and 220 calories. b. less than 200 calories.

Exercise 18: The average price of a personal computer (PC) is \$949. If the computer prices are approximately normally distributed and $\sigma = \$100$, what is the probability that a randomly selected PC costs more than \$1200? The least expensive 10% of personal computers cost less than what amount?

Exercise 19: The average length of a hospital stay for all diagnoses is 4.8 days. If we assume that the lengths of hospital stays are normally distributed with a variance of 2.1, then 10% of hospital stays are longer than how many days? Thirty percent of stays are less than how many days?



Exercise 20: The average cholesterol content of a certain brand of eggs is 215 milligrams, and the standard deviation is 15 milligrams. Assume the variable is normally distributed.

- a. If a single egg is selected, find the probability that the cholesterol content will be greater than 220 milligrams.
- b. If a sample of 25 eggs is selected, find the probability that the mean of the sample will be larger than 220 milligrams.

Exercise 21: The numbers of faculty at 32 randomly selected state-controlled colleges and universities with enrollment under 12,000 students are shown below. Use these data to estimate the mean number of faculty at all state-controlled colleges and universities with enrollment under 12,000 with 92% confidence. Assume $\sigma = 165.1$.

211	384	396	211	224	337	395	121	356
621	367	408	515	280	289	180	431	176
318	836	203	374	224	121	412	134	539
471	638	425	159	324				

Exercise 22: For each conjecture, state the null and alternative hypotheses.

- a. The average age of community college students is 24.6 years.
- b. The average income of accountants is \$51,497.
- c. The average age of attorneys is greater than 25.4 years.
- d. The average score of high school basketball games is less than 88.
- e. The average resting pulse rate of male marathon runners is less than 70 beats per minute.

Deadline & Submission: A soft copy of your assignment solution has to be submitted through the LMS latest by Sunday, December 21th, 2025.

